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RESEARCH ARTICLE

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A Bayesian phase II proof-of-concept design for clinical trials with longitudinal endpoints

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Existing phase II clinical trial designs focus on a single scalar endpoint, such as a binary, continuous, or survival endpoint. In some clinical trials, such as pain management studies, the efficacy endpoint of interest is measured longitudinally. We propose a Bayesian phase II design for such clinical trials. We model the longitudinal measurement process using Bayesian hierarchical model, where subject-specific trajectory shrinks toward the population trajectory to borrow information across subjects. The Bayesian penalized spline is used to model subject-specific and population trajectories without making strong parametric assumption on their shapes. We use the area under the curve of the trajectory as the summary of the treatment effect over time. The design takes a group sequential approach and takes into account both statistical significance and clinical relevance. Bayesian criteria is proposed to make interim and final decisions based on the evidence of statistical significance and clinical relevance. The proposed design is highly flexible and can accommodate trials with one or multiple longitudinal endpoints, as well as a longitudinal primary endpoint with a secondary endpoint. Simulation study shows that the proposed design is robust with desirable operating characteristics.

KEYWORDS

Bayesian phase II design, longitudinal endpoint, pain management, proof-of-concept design

1 | INTRODUCTION

Phase II or proof-of-concept (PoC) clinical trials play a critical gatekeeping role in developing novel drugs, of which the main purpose is to determine whether a new drug demonstrates sufficient efficacy to warrant further investigation in a confirmatory trial. Beyond classic phase II designs such as Gehan's design[1] and Simon's optimal two-stage design,[2] a number of Bayesian designs have been proposed for this purpose. Thall and Simon[3] proposed to monitor phase II trials and make go/no-go decision based on the posterior probability that the response rate of the treatment is greater than a prespecified threshold. Heitjan[4] advocated the use of a persuasion probability as a consistent criterion to determine whether a drug is promising or not. Tan et al[5] proposed Bayesian two-stage designs that mimic frequentist multistage designs. Lee et al[6] proposed to use Bayesian predictive probability to make go/no-go decisions. Cai et al[7] developed a Bayesian phase II trial design that handles delayed efficacy outcomes using multiple imputation. Zhou et al[8,9] proposed the Bayesian optimal phase II (BOP2) design as a unified approach to handle binary, ordinal, survival, and multiple endpoints.

Some methods aim to incorporate both statistical significance and clinical relevance into decision making. This motivated dual-criterion phase II designs that recommend trinary go/consider/no-go decisions, instead of traditional

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binary go/no-go decisions. Fisch et al[10] proposed a Bayesian framework for decision-making in proof-of-concept trials in settings with dual criteria. Pulkstenis et al[11] proposed a flexible Bayesian quantitative decision-making paradigm that evaluates evidence relative to achieving a multilevel target product profile. Zhao et al[12] extended BOP2 and proposed an optimal Bayesian design BOP2-DC based on dual criteria of statistical significance and clinical relevance. Most of these existing designs focus on the trials with a single scalar endpoint (eg, the response rate).

In some settings the efficacy endpoint of interest is measured longitudinally and summarized into an integrated treatment effect over the time period. One such therapeutic area is pain. The management of pain is increasingly important in treatment of many severe diseases, for example, in arthritis, headache, nerve, and advanced phases of cancer, where the aim of treatment shifts from cure to palliation.[13] U.S. Food and Drug Administration[14] (FDA) has released "Analgesic Indications: Developing Drug and Biological Products," for sponsors of analgesic products intended to treat acute, chronic and breakthrough pain. FDA also launched the Analgesic Clinical Trial Innovations, Opportunities, and Networks (ACTION) initiative, which is a collaborative effort designed to bridge gaps in the discovery and development of safe and efficacious analgesics.[15,16]

Pain clinical trials come with inherent and unique challenges. As a patient reported outcome (PRO), pain is typically measured longitudinally and highly heterogenous across subjects. This requires more complicated modeling strategy to capture the aggregated treatment effect over time and account for patient heterogeneity. In addition, several scales, including numerical rating scales (NRS), visual analog scales (VAS), and behavioral pain scales (BPS), have been proposed to measure pain. There is no consensus on which one is the best to use.[13] It is desirable to propose a design that has the flexibility to incorporate multiple scales into decision making when appropriate.

To address these challenges, we propose a Bayesian PoC design for phase II pain management clinical trials. We model longitudinal pain measurements using a Bayesian hierarchical model, where the subjective-specific pain trajectory is partitioned into the population mean trajectory and subject-specific deviation from the population trajectory. We shrink subject-specific trajectory toward the population trajectory to borrow information across subjects. The Bayesian penalized spline is used to model subject-specific and population trajectories without making strong parametric assumption on their shapes. We use the area under the curve (AUC) of the trajectory as the summary of the treatment effect over time. The design takes a group sequential approach and accounts for both statistical significance and clinical relevance. Bayesian criteria are proposed to make interim and final decisions based on the evidence of statistical significance and clinical relevance. The proposed design is highly flexible and can accommodate trials with one or multiple longitudinal endpoints, as well as a longitudinal primary endpoint with a secondary endpoint. It is important to note that we focus on using pain clinical trials to describe the methodology and illustrate its application, but the proposed design is directly applicable to other therapeutic areas involving longitudinal endpoints.

The remainder of this article is organized as follows. Section 2 proposes Bayesian hierarchical penalized spline model and the PoC trial design. Section 3 presents simulation studies to evaluate the operating characteristics of the proposed design, and Section 4 concludes this article with a brief discussion.

2 | METHOD

2.1 | Probability model

Let y_{ij} denote the jth measure of pain intensity, based on a certain scale such as NRS or VAS, for subject i, where $i = 1, \dots, I$ and $j = 1, \dots, J$. Let t_{ij} denote the time when y_{ij} is measured. We model y_{ij} as a function of t_{ij} as follows:

$$y_{ij} = f(t_{ij}) + f_i(t_{ij}) + \epsilon_{ij}, \tag{1}$$

specifically, $f(\cdot)$ is a smooth function representing the population trajectory of the pain measure over time, $f_i(\cdot)$ is a smooth function quantifying how the subject-specific trajectory deviates from the population trajectory. We assume the error term $\epsilon_{ij} \sim N\left(0, \sigma_\epsilon^2\right)$.

We model non-parametric functions $f(\cdot)$ and $f_i(\cdot)$ using penalized low-rank thin-plate splines, which has been shown to have good numerical properties.[17] Other basis functions (eg, natural cubic splines, B-splines, and radial splines), however, can also be used.[18] Specifically, $f(t_{ij})$ is represented as

$$f(t_{ij}) = \beta_0 + \beta_1 t_{ij} + \sum_{k=1}^{K} u_k |t_{ij} - \kappa_k|^3, \tag{2}$$

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where β_0, β_1 and $\boldsymbol{u} = (u_1, \ldots, u_K)$ are regression coefficients, and $\kappa_1 < \kappa_2 < \cdots < \kappa_K$ are K prespecified knots.

One attractive feature of the penalized spline is that it can be represented as a linear mixed model after appropriate reparameterization.[17,19,20] Define $\boldsymbol{Z}_K$ as the $(I \times J) \times K$ matrix, where $((i-1) \times J + j)$th row of $\boldsymbol{Z}_K$ is given by $\{|t_{ij} - \kappa_1|^3, \ldots, |t_{ij} - \kappa_K|^3\}$. Define $\boldsymbol{\Omega}_K$ as the $K \times K$ matrix, (l, k)th entry of $\boldsymbol{\Omega}_K$ is denoted as $|\kappa_l - \kappa_k|^3$. Let $z_{t_{ij}k}$ denote the $((i-1) \times J + j, k)$th entry of $\boldsymbol{Z} = \boldsymbol{Z}_K \boldsymbol{\Omega}_K^{-1/2}$, which represents the design matrix for the thin-spline random coefficients, and b_k denotes kth element of vector $\boldsymbol{b} = \boldsymbol{\Omega}_K^{1/2} \boldsymbol{u}$. For notation brevity, we suppress ij and denote t_{ij} as t in the following section. It follows that $f(\cdot)$ has the following linear mixed model representation:

$$f(t) = \beta_0 + \beta_1 t + \sum_{k=1}^{K} b_k z_{tk}, \tag{3}$$

$$b_k \sim N\left(0, \tau_b^2\right), \tag{4}$$

where β_0 and β_1 are fixed effect coefficients, and $b_1, \ldots, b_K$ are random effects coefficients, which correspond to the penalty term of the penalized spline and control the smoothness of $f(t)$. We assume non-informative priors for fixed effects $\beta_0, \beta_1 \sim N(0, 10^6)$.

Similarly, the deviation of subject-specific curve from the overall curve can also be modeled using a penalized spline and represented as a linear mixed model:

$$f_i(t) = \delta_{0i} + \delta_{1i} t + \sum_{k=1}^{K} c_{ik} z_{tk} \tag{5}$$

$$\delta_{0i} \sim N\left(0, \tau_0^2\right), \ \delta_{1i} \sim N\left(0, \tau_1^2\right), \ c_{ik} \sim N\left(0, \tau_c^2\right).$$

By centering subject-specific effects δ_{0i} and δ_{1i} at 0, this model shrinks the subjective-specific curve toward the common population curve $f(t)$, rendering information borrowing across subjects. Such information borrowing is important because it enables more reliable estimation of the pain intensity curve, for example, at the interim analysis, when some subjects have limited number of (eg, 1 or 2) observations. Unlike δ_{0i} and δ_{1i}, randomized effects c_{ik} are resulted from the random-effects-model representation of the penalized spline. They correspond the penalized term of the penalized spline, controlling the smoothness of the spline. For notational brevity, we assume that the number of knots K is the same in $f(t)$ and $f_i(t)$, but that is not required. For example, when appropriate, we can use more knots for modeling $f(t)$ than $f_i(t)$. To complete Bayesian specification of the model, we employ the following non-informative priors for model parameters: $\sigma_\epsilon^{-2}, \tau_b^{-2}, \tau_c^{-2}, \tau_0^{-2}, \tau_1^{-2} \sim Gamma(10^{-6}, 10^{-6})$.

We use the AUC of pain intensity curve as a summary of the treatment effect over the time. Let $t_{\max}$ denote the maximum observed time, and D denote the observed data, the AUC is given by

$$\text{AUC} = \int_{t=0}^{t_{\max}} \int \cdots \int f(t|\beta_0, \beta_1, b_1, \ldots, b_K) \cdot f(\beta_0|D) \cdots f(b_K|D) d\beta_0 \cdots db_K dt. \tag{6}$$

It is important to note that we here use pain management as an example of therapeutic areas, and the AUC to summarize the treatment effect. The proposed methodology is applicable to other therapeutic areas with longitudinal endpoints, for example, studies involving patient-reported outcomes such as quality of life. In addition, other than the AUC, any function of $f(t)$, say $g(f(t))$, that is deemed clinically sound can be used as the summary of the treatment effect. For example, pain intensity difference (PID) or time-weighted sum pain intensity differences (SPID), are also used to measure the treatment effect in some pain studies. Given the posterior samples of the parameters appeared in $f(t)$, the posterior of $g(f(t))$ can be easily obtained to make go/consider/no-go decisions as described later.

2.2 | Multiple endpoints

As described previously, pain has complicated and multifaceted physiology. In some settings, it is desirable to consider multiple endpoints to capture different aspects of the treatment effects for pain. For example, we may consider using two measures of pain intensity (eg, NRS and VAS) or two symptoms (eg, pain intensity and fatigue) as multiple endpoints to better characteristics of the treatment effects. Per FDA Guidance on Multiple Endpoints in Clinical Trials,[21] multiple

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endpoints may be used when (1) there are several important aspects of a disease or several ways to assess an important aspect, (2) there is no consensus about which one will best serve the study purposes, and (3) an effect on any one will be sufficient as evidence of effectiveness to support approval. These situations often occur in pain management studies. This section describes how to model M longitudinal endpoints, for example, pain intensity measured over time by using NRS and VAS. Let $y_{m,ij}$ denote the jth measurement of the mth endpoint for subject i at time t_{ij}, where $m = 1, \ldots, M$, $i = 1, \ldots, I$, and $j = 1, \ldots, J$. To simplify notations, we assume M endpoints have the same number of longitudinal measures collected at the same time grid, but this assumption is not needed for the methodology. Along the similar line as above, we model the mth endpoint $y_{m,ij}$, $m = 1, \ldots, M$, using Bayesian penalized spline model as follows:

$$y_{m,ij} = f_m(t_{ij}) + f_{mi}(t_{ij}) + \epsilon_{m,ij}, \tag{7}$$

where $\epsilon_{m,ij} \sim N(0, \sigma^2_{\epsilon m})$ is error term, $f_m(\cdot)$ denote the smooth functions for population curves, and $f_{mi}(\cdot)$ denote the deviations of subject-specific curves from the population curves, for the mth endpoint. For notation brevity, we suppress subscript ij and denote t_{ij} as t. We model $f_m(\cdot)$ and $f_{mi}(\cdot)$ using penalized low-rank thin-plate splines, represented as mixed models as follows.

$$f_m(t) = \beta_{m0} + \beta_{m1} t + \sum_{k=1}^{K} b_{mk} z_{tk}, \tag{8}$$

$$f_{mi}(t) = \delta_{m0i} + \delta_{m1i} t + \sum_{k=1}^{K} c_{mik} z_{tk}, \tag{9}$$

$$b_{mk} \sim N\left(0, \tau^2_{bm}\right) \qquad c_{mik} \sim N\left(0, \tau^2_{cm}\right), \tag{10}$$

where β_{m0} and β_{m1} are fixed effects and b_{mk} is random effect for the population curve; δ_{m0i}, δ_{m1i} and c_{mik} are randomized effects associated with subject-specific curves.

To account for potential correlation between endpoints $y_{1,ij}, \ldots, y_{M,ij}$, we assume that the random intercepts and slopes in Equation (8) follow multivariate normal distributions:

$$(\delta_{10i}, \ldots, \delta_{M0i}) \sim N_M(\mathbf{0}, \boldsymbol{D_0}), \quad (\delta_{11i}, \ldots, \delta_{M1i}) \sim N_M(\mathbf{0}, \boldsymbol{D_1}),$$

where $\mathbf{0}$ is a vector of 0 with the length of M, and $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$ are $M \times M$ covariance matrixes. We assign $\boldsymbol{D_0}^{-1}$ and $\boldsymbol{D_1}^{-1}$ independent vague Wishart prior: $\boldsymbol{D_0}^{-1}, \boldsymbol{D_1}^{-1} \sim W_M(\boldsymbol{S}, 3)$, where $\boldsymbol{S} = 10^{-4}\boldsymbol{I}_M$ and $\boldsymbol{I}_M$ denotes a $M \times M$ identify matrix. For other parameters in the model, we specify the following vague priors: $\beta_{m0}, \beta_{m1} \sim N(0, 10^6)$, $\sigma^{-2}_{\epsilon 1}, \sigma^{-2}_{\epsilon 2}, \tau^{-2}_{b1}, \tau^{-2}_{b2}, \tau^{-2}_{c1}, \tau^{-2}_{c2} \sim Gamma(10^{-6}, 10^{-6})$. As the go/consider/no-go decisions of the designs are based on the marginal estimate of the AUC associated with each endpoint (see the next section), we found that ignoring the correlation between $y_{m,ij}$'s and assigning δ_{m0i} and δ_{m1i} independent normal distribution as in Section 2.1 has negligible effects on the operating characteristics of the design. This greatly simplifies the computation and speeds up the convergence of Markov chain Monte Carlo.

2.3 | Bayesian PoC trial design

The proposed Bayesian PoC design consists of R interim looks, occurred when the number of enrolled patients reaches n_1, $n_2, \ldots, n_R$. At each look, go/consider/no-go decision is made based on the evaluation of evidence of statistical significance and clinical relevance. Specifically, let AUC_0 denote the reference AUC value, which can be the AUC estimate based on the concurrent control in a randomized trial, or based on the historical data in a single-arm trial. Let Δ denote the target clinical meaningful or relevant difference, and D_n denote the interim data with the interim sample size of n. The value of Δ should be elicited from subject matter experts based on the characteristics and objectives of the trial. Without loss of generality, we assume that a larger value of AUC means a better clinical outcome. We define evidence of significance and evidence of relevance as follows:

- *Evidence of significance*: the posterior probability that efficacy of the investigational drug is greater than the reference given the observed data, that is,

$$\Pr(\text{AUC} - \text{AUC}_0 > 0 \mid D_n).$$

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TABLE 1 Decision rule for proof-of-concept design based on evidence of significance and evidence of relevance.

		Evidence of relevance		
Analysis	**Evidence of significance**	**Strong**	**Moderate**	**Weak**
Interim	Strong	Superiority stop	Continue	Continue
	Moderate	Continue	Continue	Continue
	Weak	Continue	Continue	Futility stop
Final	Strong	Go	Consider	Consider
	Moderate	Consider	Consider	Consider
	Weak	Consider	Consider	No-go

To facilitate the establishment of PoC in interim and final analyses, we define strong, moderate, and weak evidence of significance by two probability cutoffs α and α_f as follows:

Strong evidence of significance if $\Pr(\text{AUC} - \text{AUC}_0 > 0 \mid D_n) \geq \alpha$.
- Moderate evidence of significance if $\alpha_f \leq \Pr(\text{AUC} - \text{AUC}_0 > 0 \mid D_n) < \alpha$.
- Weak evidence of significance if $\Pr(\text{AUC} - \text{AUC}_0 > 0 \mid D_n) < \alpha_f$.

Generally, α is a large probability cutoff (eg, 95%) and α_f is a probability cutoff smaller than α. In practice, the determination of α and α_f should be guided by simulation to ensure desirable operating characteristics.

- *Evidence of relevance*: the posterior probability that efficacy of the investigational drug, relative to reference, is larger than target value Δ given the observed data, that is,

$$\Pr(\text{AUC} - \text{AUC}_0 > \Delta \mid D_n).$$

Similarly, we claim

- Strong evidence of relevance if $\Pr(\text{AUC} - \text{AUC}_0 > \Delta \mid D_n) \geq \gamma$,
- Moderate evidence of significance if $\gamma_f \leq \Pr(\text{AUC} - \text{AUC}_0 > \Delta \mid D_n) < \gamma$,
- Weak evidence of significance if $\Pr(\text{AUC} - \text{AUC}_0 > \Delta \mid D_n) < \gamma_f$,

where γ is a moderate probability cutoff (eg, 50%) and γ_f is a probability cutoff smaller than γ. Noting that when $\text{AUC} = \text{AUC}_0 + \Delta$, $\Pr(\text{AUC} - \text{AUC}_0 > \Delta \mid D_n) = 0.5$, thus the value of γ is generally not too large. Both cutoffs should be calibrated by simulations to obtain desirable operating characteristics.

Given the above definition, the decision rule of the proposed Bayesian PoC design is displayed in Table 1 and described as follows:

1. At each interim,

   a) stop the trial for superiority if both strong evidence of significance and strong evidence of relevance are established,
   b) stop the trial for futility if both weak evidence of significance and weak evidence of relevance are observed,
   c) otherwise, continue the trial to the next interim.

2. At end of the trial,

   a) make a "Go" decision if both strong evidence of significance and strong evidence of relevance are established,
   b) make a "No-go" decision if both weak evidence of significance and weak evidence of relevance are observed,
   c) otherwise, make a "Consider" decision.

At the end of the trial, if "Consider" decision is made, other data, such as a secondary endpoint, may be used to provide additional information for final decision-making. For example, considering a binary secondary endpoint z with

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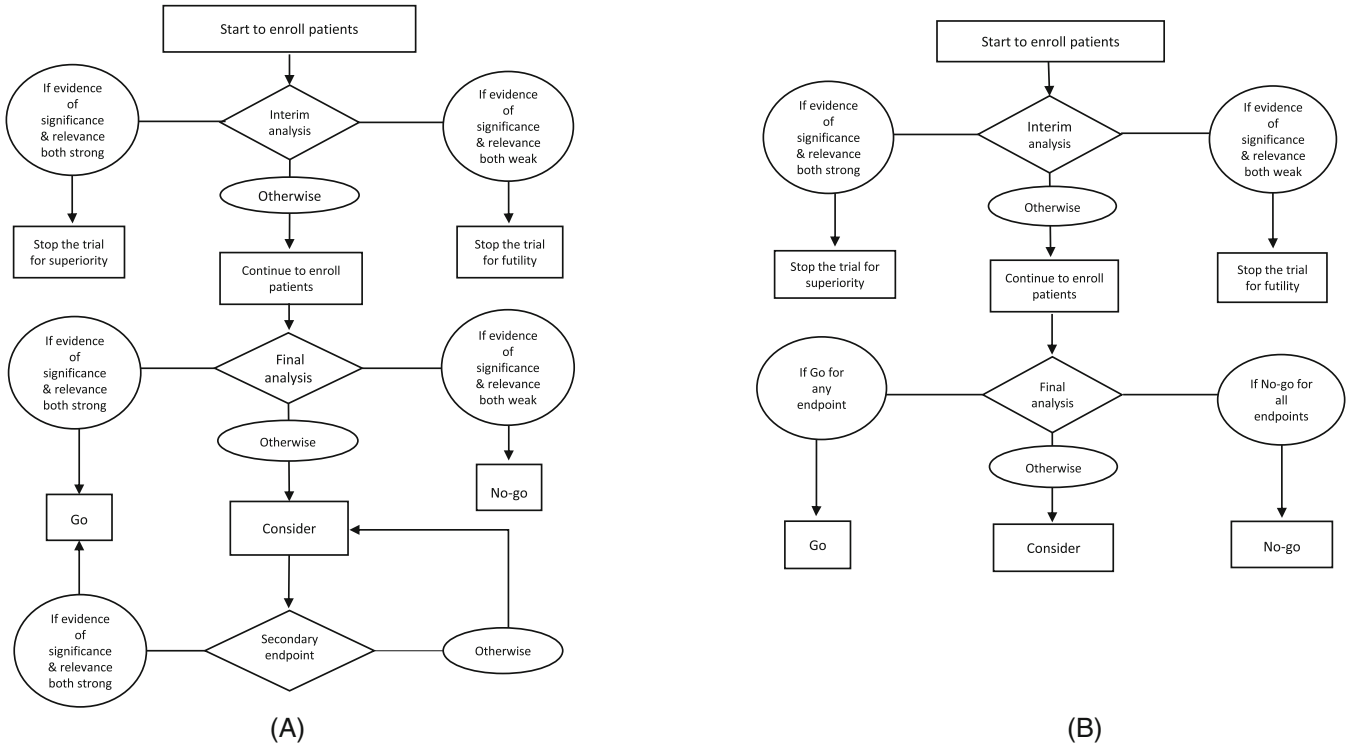


FIGURE 1 (A) Design for a single primary endpoint with a secondary endpoint, and (B) design for multiple primary endpoints.

the response rate of π. When the decision based on the primary endpoint falls in the considerate zone, we may make a "Go" decision if the following conditions are both satisfied:

$$\Pr(\pi - \pi_0 > 0 \mid D_n) \geq \alpha_z \quad and \quad \Pr(\pi - \pi_0 > \Delta_z \mid D_n) \geq \gamma_z, \tag{11}$$

where π_0 is the reference response rate of efficacy, Δ_z is the target clinical relevant difference, and α_z and γ_z are probability cutoffs. The evaluation of the above Bayesian posterior criteria can be based on a standard beta-binomial model.

The above decision-making rule and design can be readily extended to trials with M endpoints, as follows.

1. At each interim, evaluate the evidence of significance and evidence of relevance to each of the endpoints, and then

   a) stop for superiority or futility if the superiority/futility stopping rule is met for all M endpoints,
   b) otherwise, continue the trial to the next interim.

2. At end of the trial,

   a) make a "Go" decision if both strong evidence of significance and strong evidence of relevance are established for any endpoints,
   b) make a "No-go" decision if both weak evidence of significance and weak evidence of relevance are observed for all M endpoints,
   c) otherwise, make a "Consider" decision.

The choice of the number of interims should be based on both statistical and logistic considerations. More interims enable earlier go/no-go decisions for a very effective/futile treatment, at the cost of increased operational burden. The number of interims should be chosen to reach an appropriate balance between these competing considerations and calibrated using simulation to fit the purpose of the trial under consideration. In practice, 1 or 2 interims often provide a good balance in quickly identifying effective or futile treatment and logistic burden. The PoC design for a primary longitudinal endpoint with a secondary binary endpoint, and the design for two equally-important longitudinal endpoints are illustrated in Figure 1.

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3 | SIMULATION STUDIES

We conducted simulation studies to evaluate the operating characteristics of the proposed design. We generate longitudinal data y_{ij} for subject i at time $t_{ij}, j = 1, \ldots, 11$, based on a polynomial model $y_{ij} = 0.5 + 0.1t_{ij} + 0.015t_{ij}^2 + r_i + \omega_{ij}$ where $t_{ij} = (0, 1, \ldots, 10)$, $r_i \sim N(0, 0.1^2)$ is the random effect, and $\omega_{ij} \sim N(0, 0.05^2)$ is the error term. The target clinical meaningful difference $\Delta = 0.1$. Interim analyses were carried out after every 9 patients were enrolled, with a maximum sample size of 36. We calibrated the probability cutoffs α, α_f, γ, γ_f, α_z, and γ_z to control the "Go" rate under the null hypothesis (ie, false go rate) to be less than or around 10%, the "Go" rate under the alternative hypothesis (eg, true go rate) to be around 90%, and a reasonable early stop rate (ie, control the false futility stop rate to be around 5% under the alternative hypothesis). The values of probability cutoffs are specified in Tables 2–4. We considered different true treatment effects (ie, AUC), and conducted 1000 simulated trials under each scenario. We reported the the percentages of "Go," "No-go," and "Consider" decisions, the percentages of early superiority stop (PESS), the percentages of early futility stop (PEFS), and the average sample size.

Table 2a shows the results of simulated trials with a single AUC endpoint. Under the null hypothesis of AUC = 5, probability of "Go" is 10.4%, probability of "No-go" is 67.9%, and probability of "Consider" decision

TABLE 2 Percentages of go/consider/no-go rates, PESS, PEFS, and the average sample size in the proposed Bayesian PoC design.

Scenario	Endpoints	Go	No-go	Consider	PESS	PEFS	Sample size
(a) One primary longitudinal endpoint							
1	5[a]	0.104	0.679	0.217	0.092	0.647	18.88
2	5.1	0.249	0.491	0.260	0.207	0.474	21.02
3	5.2	0.406	0.325	0.269	0.336	0.317	22.20
4	5.3	0.577	0.210	0.213	0.510	0.203	21.46
5	5.4	0.788	0.115	0.097	0.693	0.112	19.56
6	5.5[b]	0.905	0.050	0.045	0.847	0.05	17.63
(b) Two primary longitudinal endpoints							
1	(5.0, 8.0)[a]	0.109	0.812	0.079	0.013	0.759	18.58
2	(5.0, 8.2)	0.320	0.596	0.084	0.035	0.559	22.87
3	(5.1, 8.2)	0.397	0.500	0.103	0.088	0.457	24.22
4	(5.2, 8.3)	0.654	0.262	0.084	0.222	0.249	26.74
5	(5.3, 8.3)	0.759	0.188	0.053	0.317	0.179	26.89
6	(5.4, 8.4)	0.894	0.095	0.011	0.524	0.095	24.88
7	(5.5, 8.5)[b]	0.959	0.035	0.006	0.755	0.035	21.74
(c) One primary longitudinal endpoint and one secondary binary endpoint							
1	(5, 0.4)[a]	0.106	0.887	0.007	0.082	0.643	19.40
2	(5.1, 0.5)	0.270	0.691	0.039	0.146	0.491	22.01
3	(5.2, 0.5)	0.450	0.510	0.040	0.292	0.323	22.65
4	(5.3, 0.6)	0.730	0.245	0.025	0.453	0.214	22.28
5	(5.4, 0.6)	0.865	0.116	0.019	0.649	0.096	21.29
6	(5.5, 0.6)[b]	0.914	0.079	0.007	0.767	0.066	19.31

Note: Probability cutoffs: (a) $\alpha = 0.95, \gamma = 0.7, \alpha_f = 0.45, \gamma_f = 0.55$, (b) $\alpha = 0.93, \gamma = 0.6, \alpha_f = 0.88, \gamma_f = 0.58$, (c) $\alpha = 0.96, \gamma = 0.7, \alpha_f = 0.45, \gamma_f = 0.55, \alpha_z = 0.96, \gamma_z = 0.6$.

[a]Null hypothesis.

[b]Alternative hypothesis.

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TABLE 3 Results of sensitivity analysis 1: Percentages of go/consider/no-go rates, PESS, PEFS, and the average sample size in the proposed Bayesian PoC design.

Scenario	Endpoints	Go	No-Go	Consider	PESS	PEFS	Sample size
(a) One primary longitudinal endpoint							
1	5[a]	0.114	0.677	0.209	0.103	0.642	19.03
2	5.1	0.269	0.494	0.237	0.239	0.468	20.18
3	5.2	0.413	0.326	0.261	0.352	0.317	21.65
4	5.3	0.597	0.212	0.191	0.530	0.207	20.73
5	5.4	0.790	0.119	0.091	0.712	0.118	19.01
6	5.5[b]	0.901	0.059	0.050	0.855	0.059	16.59
(b) Two primary longitudinal endpoints							
1	(5.0, 8.0)[a]	0.104	0.820	0.076	0.008	0.767	18.43
2	(5.0, 8.2)	0.317	0.595	0.088	0.037	0.558	22.87
3	(5.1, 8.2)	0.412	0.487	0.101	0.086	0.450	24.40
4	(5.2, 8.3)	0.667	0.259	0.074	0.228	0.247	26.74
5	(5.3, 8.3)	0.764	0.195	0.041	0.327	0.183	26.47
6	(5.4, 8.4)	0.890	0.094	0.016	0.539	0.092	24.80
7	(5.5, 8.5)[b]	0.958	0.038	0.004	0.754	0.038	21.68
(c) One primary longitudinal endpoint and one secondary binary endpoint							
1	(5, 0.4)[a]	0.110	0.883	0.007	0.090	0.644	19.20
2	(5.1, 0.5)	0.296	0.672	0.032	0.170	0.478	21.88
3	(5.2, 0.5)	0.477	0.487	0.036	0.318	0.308	22.60
4	(5.3, 0.6)	0.721	0.257	0.022	0.471	0.223	21.60
5	(5.4, 0.6)	0.864	0.115	0.021	0.661	0.100	20.76
6	(5.5, 0.6)[b]	0.919	0.076	0.005	0.788	0.069	18.43

Note: Probability cutoffs: (a) $\alpha = 0.945, \gamma = 0.7, \alpha_f = 0.45, \gamma_f = 0.55$, (b) $\alpha = 0.93, \gamma = 0.6, \alpha_f = 0.88, \gamma_f = 0.58$, (c) $\alpha = 0.955, \gamma = 0.7, \alpha_f = 0.45, \gamma_f = 0.55, \alpha_z = 0.96, \gamma_z = 0.6$.

[a]Null hypothesis.

[b]Alternative hypothesis.

is 21.7%. The probability of "Go" increases with the increase of true AUC values from 10.4% to 90.5%, the probability of "No-go" decreases from 67.9% to 5.0%, and probability of "Consider" changes from 21.7% to 4.5% in scenarios 1-6.

Table 2b shows the results of simulated trials with two primary endpoints AUC_1 and AUC_2, where the values of AUC_1 were kept as the same as those in the case of a single endpoint. Comparing to the results in one longitudinal endpoint, considering two primary endpoints can improve the efficiency of the design in some scenarios. For example, in scenario 2, the probability of "Go" increases from 10.4% (scenario 1 in Table 2a) to 32.0%, the probability of "No-go" decreases from 67.9% to 59.6%. In scenario 4, the probability of "Go" increases to 65.4% compared to 40.6% in scenario 3 of Table 2a, and probability of "No-go" decreases to 26.2% compared to 32.5%. Same trends can be found in all other scenarios in Table 2b.

Table 2c shows the simulation results with a primary endpoint and a secondary endpoint, where the secondary endpoint was used to make final decision when the decision based on the primary endpoint is "Consider." Compared to the results in one primary longitudinal endpoint, the probability of "Consider" decision dramatically decreases with the inclusion of a secondary endpoint. For example, in scenario 1, the probability of "No-go" changes from 67.9% to 88.7%, and the probability of "Consider" decreases from 21.7% to 0.7%. In scenario 5, the probability of "Go" increases from 78.8% to 86.5%, and the probability of "Consider" decreases from 9.7% to 1.9%. All other scenarios show the same trend.

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TABLE 4 Results of sensitivity analysis 2: Percentages of go/consider/no-go rates, PESS, PEFS and actual sample size in the proposed Bayesian PoC design.

Scenario	Endpoints	Go	No-Go	Consider	PESS	PEFS	Sample size
(a) One primary longitudinal endpoint							
1	5[a]	0.113	0.683	0.204	0.097	0.652	18.53
2	5.1	0.254	0.514	0.232	0.208	0.489	20.18
3	5.2	0.401	0.342	0.257	0.343	0.333	21.59
4	5.3	0.592	0.224	0.184	0.514	0.216	20.59
5	5.4	0.789	0.122	0.089	0.704	0.121	19.12
6	5.5[b]	0.901	0.064	0.035	0.844	0.064	16.95
(b) Two primary longitudinal endpoints							
1	(5.0, 8.0)[a]	0.097	0.811	0.092	0.009	0.757	18.59
2	(5.0, 8.2)	0.328	0.587	0.085	0.034	0.555	23.03
3	(5.1, 8.2)	0.407	0.492	0.101	0.079	0.449	24.60
4	(5.2, 8.3)	0.667	0.259	0.074	0.235	0.249	26.73
5	(5.3, 8.3)	0.761	0.189	0.050	0.335	0.180	26.47
6	(5.4, 8.4)	0.891	0.094	0.015	0.523	0.093	24.55
7	(5.5, 8.5)[b]	0.953	0.039	0.008	0.744	0.039	21.73
(c) One primary longitudinal endpoint and one secondary binary endpoint							
1	(5, 0.4)[a]	0.109	0.884	0.007	0.092	0.663	18.62
2	(5.1, 0.5)	0.280	0.688	0.032	0.156	0.502	21.49
3	(5.2, 0.5)	0.444	0.516	0.040	0.284	0.339	22.50
4	(5.3, 0.6)	0.723	0.262	0.015	0.468	0.228	21.38
5	(5.4, 0.6)	0.852	0.131	0.017	0.664	0.118	20.30
6	(5.5, 0.6)[b]	0.906	0.089	0.005	0.781	0.076	18.60

Note: Probability cutoffs: (a) $\alpha = 0.94, \gamma = 0.7, \alpha_f = 0.45, \gamma_f = 0.55$, (b) $\alpha = 0.93, \gamma = 0.6, \alpha_f = 0.88, \gamma_f = 0.58$, (c) $\alpha = 0.95, \gamma = 0.7, \alpha_f = 0.45, \gamma_f = 0.55, \alpha_z = 0.96, \gamma_z = 0.6$.

[a]Null hypothesis.

[b]Alternative hypothesis.

4 | AN ILLUSTRATIVE TRIAL EXAMPLE

Consider a single-arm phase II clinical trial that studies the efficacy of tumescent anesthesia in pain management during a dynamic phototherapy session performed for the management of actinic keratosis. Patients are followed up for eight consecutive weeks, score of the maximum numerical pain assessment (scale from 0 to 10) felt during the therapy session is measured every week. The primary endpoint is the AUC of the pain assessment measure scores. The current standard treatment is paracetamol, which provides a reference AUC value of 54. The total sample size is 30, and one interim analysis is taken when 15 patients are enrolled. The probability cutoffs α, α_f, γ, and γ_f are calibrated by having the false go rate to be around 10% under the null, that is, H_0 : AUC = 54; the true go rate to be around 90% and the false futility stop rate to be around 5% under the alternative analysis, that is, H_0 : AUC = 57. The target clinical meaningful difference $\Delta = 1$. Estimated AUC with the collected data is 56.05, posterior probabilities $\Pr(\text{AUC} - \text{AUC}_0 > 0 \mid D_n) = 0.938$ and $\Pr(\text{AUC} - \text{AUC}_0 > \Delta \mid D_n) = 0.788$, thus both strong evidence of significance and evidence of relevance are observed. According to the decision rules of the PoC design, superiority stop is made at interim analysis.

5 | SENSITIVITY ANALYSES

We conduct sensitivity analyses to evaluate the robustness of proposed PoC design. Table 3 presents simulation results using different priors for $\sigma_\epsilon^{-2}, \tau_b^{-2}, \tau_c^{-2}, \tau_0^{-2}, \tau_1^{-2} \sim Gamma(10^{-9}, 10^{-9})$. Table 4 presents simulation results when y_{ij} were

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generated under another model $y_{ij} = 0.04 + 0.004t_{ij} + 0.0006t_{ij}^2 + 0.00008t_{ij}^3 + r_i + \omega_i$. Results show similar patterns as those in Table 2, suggesting that the proposed method is robust.

6 | DISCUSSION

We proposed a Bayesian phase II PoC design for clinical trials with longitudinal endpoints. The proposed design models longitudinal endpoints using flexible penalized spline, and accounts for patient heterogeneity by specifying the population mean trajectory, as well as subject-specific trajectories, through a Bayesian hierarchical model. This design takes a group sequential approach and takes into account both statistical significance and clinical relevance for go/consider/no-go decision-making. The proposed methodology is highly flexible and capable of accommodating a single endpoint and multiple endpoints. Extensive simulation studies show that the proposed design generates robust and desirable operating characteristics.

One important feature of our design is that it automatically handles missing data and dropouts owing to its mixed model presentation and information borrowing across subjects via random effects (or the hierarchical model structure from the Bayesian perspective).[22] Specifically, when some subjects have less than J measures due to missing visits or dropout, the model and design are still directly applicable based on the observed data, assuming that the missing data are missing at random. When missing data are not missing at random, the longitudinal model can be extended to include the missing data mechanism to adjust for the nonignorable missing data. This could be a topic of future research. The proposed model can be extended to randomized controlled trials with the inclusion of a treatment arm as covariate in the Bayesian hierarchical model. When appropriate, other covariates can also be added to the model to adjust for confounders.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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